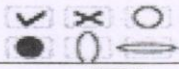


天津工业大学 (2019-2020 学年第一学期)

《概率论与数理统计》理工期末试卷 (A 卷) (2020.1 数学科学学院)

姓名:		学 号									
学院:											
班级:											
正确填涂: ■											
错误填涂											
											
缺考 ☐											
违纪 ☐											

一. 填空题 (每题 2 分, 共 36 分)

1. ① <u>0.15</u> , ② <u>0.75</u> , ③ <u>$\frac{7}{12}$</u> .	4. ① <u>$\frac{1}{2}$</u> , ② <u>$\frac{13}{16}$</u> ,
2. ① <u>$\frac{1}{30}$</u> , ② <u>$\frac{1}{9}$</u>	5. ① <u>$1-3e^{-2}$</u> , ② <u>e^{-5}</u> ,
3. ① <u>$f(x) = \frac{1}{2\sqrt{\pi}} e^{-\frac{(x-1)^2}{8}}$ $-\infty < x < +\infty$</u>	③ <u>1</u> , ④ <u>39</u> .
② <u>$\frac{1}{2}$</u> , ③ <u>5</u> .	6. ① <u>$\frac{1}{9}$</u> , ② <u>6</u> ,
	③ <u>$F(6, 6)$</u> , ④ <u>$t(6)$</u> .

非填空题

二. (10分).

请在各题目的答题区域内作答, 超出黑色矩形边框限定区域的答案无效

二. 设 A : 甲同学上学迟到, B_1 : 乘地铁上学, B_2 : 乘公共汽车上学

则: 1.
$$P(A) = P(A|B_1)P(B_1) + P(A|B_2)P(B_2)$$
$$= 0.05 \times 0.7 + 0.1 \times 0.3 = 0.065$$

2.
$$P(B_2|A) = \frac{P(A|B_2)P(B_2)}{P(A)} = \frac{0.1 \times 0.3}{0.065} = \frac{6}{13}$$

三. 1.
$$P\{X > 3\} = \int_3^{+\infty} f(x) dx = \int_3^{+\infty} 2^{-x} \ln 2 dx = \frac{1}{8}$$

2.
$$P\{Y = n\} = C_{n-1}^1 \left(\frac{1}{8}\right)^2 \left(\frac{7}{8}\right)^{n-2}, \quad n = 2, 3, \dots$$

四.
$$X = \sum_{k=1}^{400} X_k \xrightarrow{\text{近似}} N(440, 8^2)$$

$$\therefore P\{X \leq 464\} = P\left\{\frac{X-440}{8} \leq \frac{464-440}{8}\right\}$$

$$\approx \Phi(3) \approx 0.9987$$

五. 1.
$$\int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f(x, y) dx dy = 1$$

即:
$$\int_0^2 dx \int_0^2 kx e^y dy = 1$$
 . 解得: $k = \frac{1}{2(e^2-1)}$

2.
$$f_Y(y) = \int_{-\infty}^{+\infty} f(x, y) dx = \begin{cases} \int_0^2 \frac{1}{2(e^2-1)} x e^y dx, & 0 < y < 2 \\ 0, & \text{其它} \end{cases}$$

$$= \begin{cases} \frac{e^y}{e^2-1}, & 0 < y < 2 \\ 0, & \text{其它} \end{cases}$$

$$3. 0 < y < 2 \text{ 时, } f_{X|Y}(x|y) = \frac{f(x,y)}{f_Y(y)} = \begin{cases} \frac{x}{2}, & 0 < x < 2 \\ 0, & \text{其它} \end{cases}$$

$$4. E(2Y+1) = 2E(Y) + 1 = 2 \int_{-\infty}^{+\infty} y f_Y(y) dy + 1 \\ = 2 \int_0^2 y \cdot \frac{e^y}{e^2-1} dy + 1 = \frac{2(e^2+1)}{e^2-1} + 1 = \frac{3e^2+1}{e^2-1}$$

六. 1.

X	-1	0	1
P _k	0.3	0.3	0.4

Y	2	4
P _k	0.4	0.6

2.

X	-1	0	1
P{X=k Y=4}	$\frac{1}{3}$	$\frac{1}{6}$	$\frac{1}{2}$

, $P\{X=k|Y=4\} = \frac{P\{X=k, Y=4\}}{P\{Y=4\}}$
k = -1, 0, 1

3. $D(X) = E(X^2) - (E(X))^2 = 0.7 - (0.1)^2 = 0.69$

$D(Y) = E(Y^2) - (E(Y))^2 = 11.2 - (3.2)^2 = 0.96$

4. $\text{Cov}(X, Y) = E(XY) - E(X)E(Y)$
 $= 0.4 - 0.32 = 0.08$

七. 1. $\bar{X} = \frac{\sum_{i=1}^n X_i}{n}$, $E(X) = \int_{-\infty}^{+\infty} x f(x) dx = \int_0^1 x \cdot \sqrt{\theta} x^{\sqrt{\theta}-1} dx$
 $= \frac{\sqrt{\theta}}{\sqrt{\theta}+1}$

令 $E(X) = \bar{X}$. 即: $\frac{\sqrt{\theta}}{\sqrt{\theta}+1} = \bar{X}$. 得 $\hat{\theta}_1 = \left(\frac{\bar{X}}{1-\bar{X}}\right)^2$

2. $L(\theta) = \prod_{i=1}^n (\sqrt{\theta} X_i^{\sqrt{\theta}-1}) = \theta^{\frac{n}{2}} \prod_{i=1}^n X_i^{\sqrt{\theta}-1}$

$\ln L(\theta) = \frac{n}{2} \ln \theta + (\sqrt{\theta}-1) \sum_{i=1}^n \ln X_i$

令 $\frac{d \ln L(\theta)}{d\theta} = \frac{n}{2\theta} + \frac{\sum_{i=1}^n \ln X_i}{2\sqrt{\theta}} = 0$

解得 $\theta = \frac{n^2}{(\sum_{i=1}^n \ln X_i)^2}$. 故 $\hat{\theta}_2 = \frac{n^2}{(\sum_{i=1}^n \ln X_i)^2}$